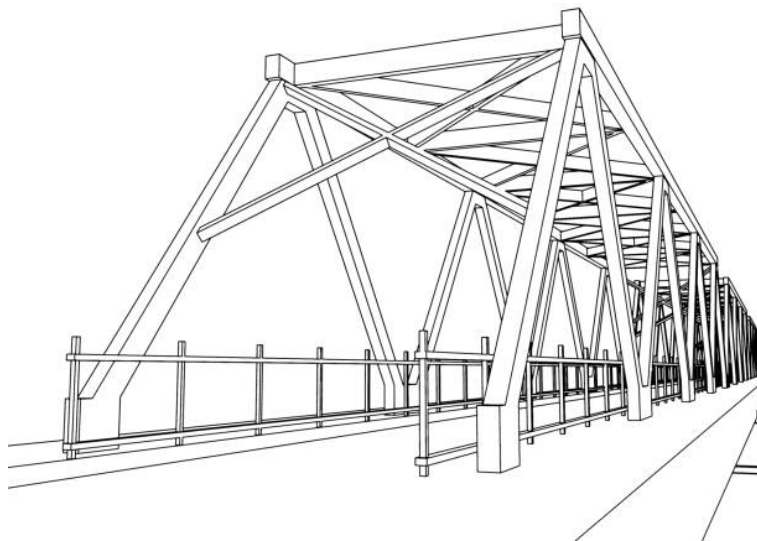




ME-473
MINI-PROJECT 2

Design of a 2D truss bridge



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1 Introduction

This project aims to design a 2D bridge spanning a 20 m gap using trusses, assuming frictionless nodes, considering only tension and compression load inside the trusses, and solving them using finite element methods. The first 5 nodes are given along with a parameter a for the height of the fifth node. The constraints on the design are as follows:

- Vertical point load of 50'000 N on the fifth node
- Maximum displacement of the nodes of 40 mm
- First natural frequency must be above 10 Hz (bonus if it is above 50 Hz)
- A threshold volume is defined for each material
- Keeping the trusses in the linear elastic range

2 Methodology

2.1 Material and physical properties

Material	Density [kg/m^3]	E [GPa]	Yield strength [MPa]
Structural steel	7850	200	215 ¹
Aluminum alloy	2700	69	110 ²
Engineered wood	600	12	30 ³

Table 1: List of the three different materials and their properties.

Yield strength for isotropic or eventually orthotropic material is, in theory, the same; however, most of the time, materials are stronger when compressed.

Structural steel comes in many different types, such as carbon steel, alloy steels, and stainless steels, as well as being available in many different tempers. For this application, 304 stainless steel was selected thanks to its corrosion resistant properties and wide use in bridge constructions. This steel has yield strengths ranging anywhere from 205 MPa to over 800 MPa when cold-worked. The T300 SS is at the lower end of this range and is selected for this project.

The yield strength of aluminum alloys highly depends on the alloying elements combined with the aluminum and the type of heat treatment (or temper) applied, despite all having similar values of Young's modulus. Yield strengths can range from approximately 24 MPa for untreated 1100 aluminum to around 483 MPa for 7075 aluminum with a T6 temper. In

¹<https://asm.matweb.com/search/specificmaterial.asp?bassnum=mq304a>

²https://www.engineeringtoolbox.com/properties-aluminum-pipe-d_1340.html

³https://www.fpl.fs.usda.gov/documnts/fplgtr/fpl_gtr190.pdf

this report, the yield strength selected corresponds to 6061 aluminum alloy, a widely used material in applications such as architectural support structures and vehicle chassis, due to its well-balanced combination of strength, corrosion resistance, and weldability.

For the engineered wood, values vary a lot; however, looking at common species of dried wood, the compression tensile strength along the grain ranges from around 30 to 50 MPa. The lower value of the range is taken for conservatism.

2.2 Governing equation and boundary conditions

The strong form of the equations of motion that govern the dynamics of the 2D truss bridge are as follows:

$$\begin{cases} EA \partial_{x'}^2 u(x', t) = \rho A \ddot{u}(x', t) \\ EA \partial_{x'} u(0, t) = -f_1(t) \\ EA \partial_{x'} u(\ell, t) = f_2(t) \end{cases} \quad \forall x' \in [0, \ell], \quad \forall t \in \mathbb{R}^+ \quad (1)$$

where x' is the local coordinate of the bar element and ℓ , its length.

The weak form is obtained by multiplying the first part of Eq.1 by the virtual displacement δu and integrating by parts. Note that the Neumann boundary conditions were taken into account.

$$\int_0^\ell EA \partial_{x'} u \partial_{x'} \delta u dx' + \int_0^\ell \rho A \ddot{u} \delta u dx' - f \delta u = 0 \quad \forall \delta u \in \mathcal{V} \quad (2)$$

where the displacement $u \in \mathcal{U}$ and the virtual displacement $\delta u \in \mathcal{V}$ belong to function spaces defined as

$$\begin{aligned} \mathcal{U} &= \{u(x', t) \in H^1(]0, \ell[) \mid \partial_{x'} u(0, t) = \frac{-f_1}{EA}, \partial_{x'} u(\ell, t) = \frac{f_2}{EA}, \forall x' \in [0, \ell], \forall t \in \mathbb{R}^+\} \\ \mathcal{V} &= \{\delta u(x') \in H^1(]0, \ell[), \forall x' \in [0, \ell]\} \end{aligned} \quad (3)$$

2.3 Finite element method (FEM)

Using the Galerkin method, we can approximate u and δu with u^h and δu^h belonging to smaller function spaces $\mathcal{U}^h$ and $\mathcal{V}^h$ respectively. FEM is then used to move on to the local method where the functions u^h and δu^h are equal to a combination of arbitrary local shape functions and their respective contributions.

$$\begin{aligned}
u(x', t) &\approx u^h(x', t) \in \mathcal{U}^h \subset \mathcal{U} \\
u^h(x', t) &= H(x') q(t) \\
\delta u(x') &\approx \delta u^h(x') \in \mathcal{V}^h \subset \mathcal{V} \\
\delta u^h(x') &= H(x') \delta q(x')
\end{aligned} \tag{4}$$

where $H(x')$ is the 1×2 vector containing the two linear shape functions and $q(t)$ is a 2×1 vector representing the displacements of the nodes. The vector $H(x')$ is defined as

$$H(x') = (h_1(x'), h_2(x')), \quad h_1(x') = 1 - \frac{x'}{\ell}, \quad h_2(x') = \frac{x'}{\ell} \quad \forall x' \in [0, \ell] \tag{5}$$

We can then derive the local semi-discrete weak formulation of the first part of Eq.1.

$$\delta \mathbf{q}^T (\mathbf{M}_{loc} \ddot{\mathbf{q}}_{loc}(t) + \mathbf{K}_{loc} \mathbf{q}_{loc}(t) - \mathbf{f}_{loc}(t)) = 0 \tag{6}$$

with

$$\begin{aligned}
\mathbf{M}_{loc} &= \int_0^\ell \rho A H(x')^T H(x') dx' = \frac{\rho A \ell}{6} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \\
\mathbf{K}_{loc} &= \int_0^\ell EA \left(\frac{dH(x')}{dx'} \right)^T \left(\frac{dH(x')}{dx'} \right) dx' = \frac{EA}{\ell} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \\
\mathbf{f}_{loc} &= \begin{pmatrix} f_1(t) \\ f_2(t) \end{pmatrix}
\end{aligned}$$

Before starting to assemble, these objects must be converted from the local domain to the global domain with a referential change. This change of reference translates to a 2D rotation around the origin of the global referential from the 1D local coordinate x' to the 2D global coordinates x and y . The new mass and stiffness matrices, and the force vector become

$${}^e\mathbf{M} = {}^eL^T \mathbf{M}_{loc} {}^eL, \quad {}^e\mathbf{K} = {}^eL^T \mathbf{K}_{loc} {}^eL, \quad {}^e\mathbf{f} = {}^eL^T \mathbf{f}_{loc} \tag{7}$$

where eL is the coordinate transformation matrix defined as

$${}^eL = \begin{pmatrix} \cos({}^e\theta) & \sin({}^e\theta) & 0 & 0 \\ 0 & 0 & \cos({}^e\theta) & \sin({}^e\theta) \end{pmatrix} \tag{8}$$

The global mass and stiffness matrices $\mathbf{M}$ and $\mathbf{K}$ as well as the global force vector $\mathbf{f}$ can now be properly assembled by adding the contribution of each node to its correct location with the connectivity table.

$$\mathbf{M} = \bigcup_{e=1}^n {}^e\mathbf{M}, \quad \mathbf{K} = \bigcup_{e=1}^n {}^e\mathbf{K}, \quad \mathbf{f} = \bigcup_{e=1}^n {}^e\mathbf{f} \tag{9}$$

where n is the number of nodes inside the 2D truss. Note that the dimensions are $2n \times 2n$ for $\mathbf{M}$ and $\mathbf{K}$, and $2n \times 1$ for $\mathbf{f}$.

It is interesting to observe that in this study the first 8 rows and columns will be eliminated from $\mathbf{M}$ and $\mathbf{K}$ (first 8 rows for $\mathbf{f}$) since the first four nodes are fixed in the 2D truss structure.

Finally, the first $2n - 8$ natural frequencies and mode shapes can be retrieved by solving the following eigenvalue problem

$$(\mathbf{K}_{\text{red}} - 4\pi^2 f^2 \mathbf{M}_{\text{red}}) \phi = 0 \quad (10)$$

where $\mathbf{M}_{\text{red}}$ and $\mathbf{K}_{\text{red}}$ are the reduced mass and stiffness matrices respectively, f the vector containing the first $2n - 8$ natural frequencies, and ϕ the modal vector.

As for the static analysis, only the global stiffness matrix $\mathbf{K}$ and global force vector $\mathbf{f}$ are required to find the final displacement of each node. They are obtained by solving the following equation

$$\mathbf{K}_{\text{red}} \Delta \mathbf{x} = \mathbf{f} \quad (11)$$

where $\Delta \mathbf{x}$ is a $(2n-8) \times 1$ matrix structured as $\Delta \mathbf{x} = (\Delta x_5, \Delta y_5, \Delta x_6, \Delta y_6, \dots, \Delta x_n, \Delta y_n)^T$, the displacement in x and y of each node.

The nodal displacements are used to calculate axial stresses in each bar using the equation

$${}^e\sigma = \frac{E}{{}^e\ell} [-\cos({}^e\theta), -\sin({}^e\theta), \cos({}^e\theta), \sin({}^e\theta)] \begin{bmatrix} q_{ix} \\ q_{iy} \\ q_{jx} \\ q_{jy} \end{bmatrix} \quad (12)$$

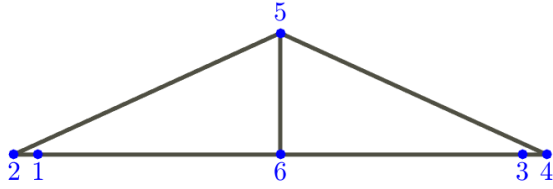
where e indicates an element by element calculation, ℓ is the length of the bar, θ is the angle of the bar, and q is the nodal displacement in the x or y direction, which bar e connected on each end to nodes i and j .

3 Results and Discussion

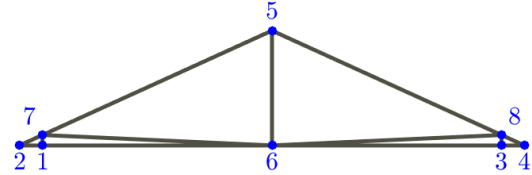
3.1 Case 1: Structural steel

Out of the numerous truss types which exist, Howe trusses were investigated for the steel bridge application as it is a commonly used shape that is known to be best for horizontal spans with vertically applied loads. Each bridge design was created with two variable parameters: truss complexity (c) and maximum bridge height (h). Truss complexity refers to the number of vertical/diagonal member triplets within the triangular main frame. The first set of internal members is placed at predefined inner pin locations, and additional

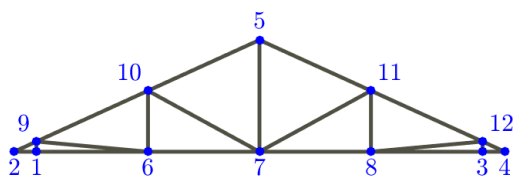
triangular sections are evenly distributed along the span. Figure 1 illustrates the node numbering and overall bridge shape schematic for complexities from zero to three. Bridge heights ranging from 0.1 to 25 m and complexities from 0 to 10 were analyzed. A cross-sectional area of 0.0050 m^2 is used for all calculations as the volume threshold was not a limiting factor in the bridge selection.



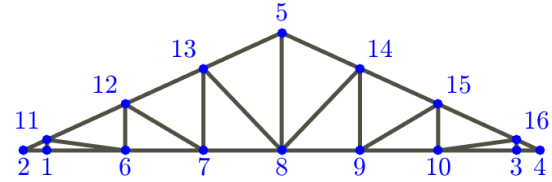
(a) Truss with complexity 0.



(b) Truss with complexity 1.



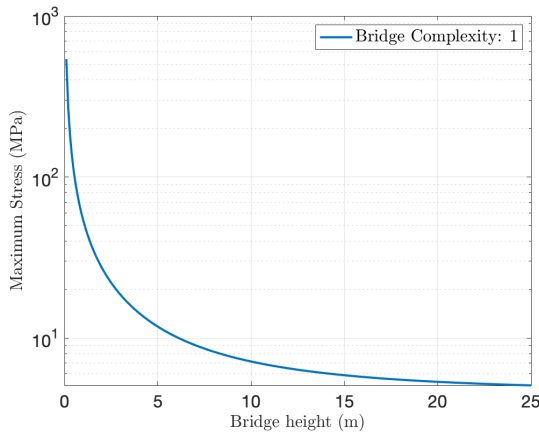
(c) Truss with complexity 2.



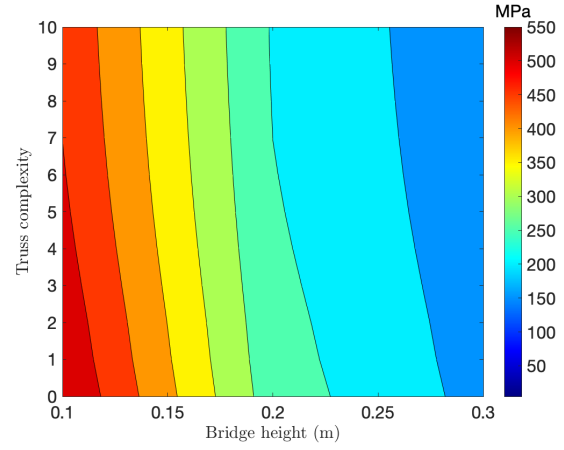
(d) Truss with complexity 3.

Figure 1: The first four complexity shapes of the Howe truss, with nodes labeled.

A summary of maximum stress variations across different bridge heights and complexities is shown in Figure 2. While increasing truss complexity reduces maximum stress, the height of the bridge has a much more significant impact on the magnitude of the stress overall. Increasing the bridge height from 0.1 m to 0.4 m reduces the maximum stress by a factor of 4, while increasing the truss complexity from 1 to 10 only decreases the maximum stress by 10%. The lowest overall stress occurs at a bridge complexity of 1 and a height of 25 with value $\sigma = 5.03 \text{ MPa}$. In order to maintain a maximum stress below the yield strength of 304 stainless steel, a bridge height of at least 0.3 m is required.



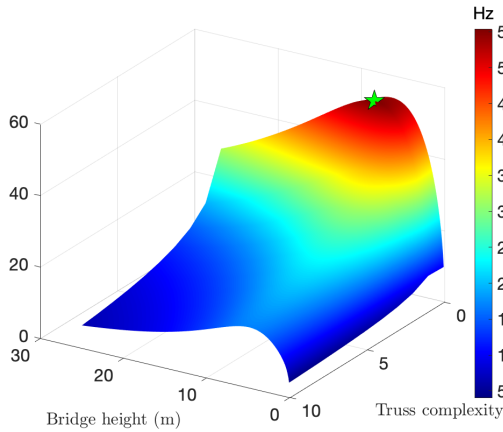
(a) Relationship for complexity = 1.



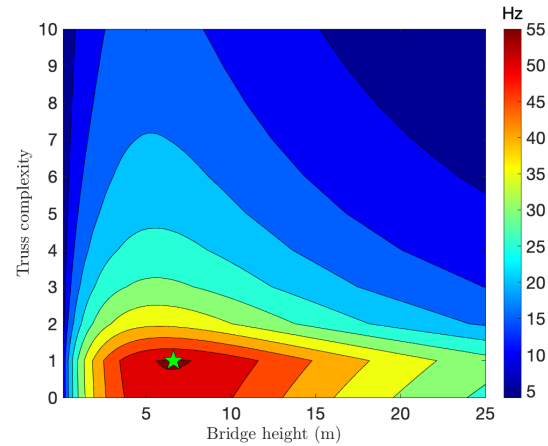
(b) Zoomed in plot of the relationship.

Figure 2: Relationship between maximum stress in any member of the truss, the height h and the complexity of the truss c .

As expected, the volume of the bridge increases with both height and complexity (Figure 5) for a set cross-sectional area of 0.0050 m^2 . In order to remain within the overall volume bounds, a tradeoff between height and complexity must be fulfilled, as shown by the iso-lines in the volume contour plot. The final selected bridge design (represented by the green star) falls well within the 1 m^3 prescribed volume limits.



(a) 3D plot of the relationship.



(b) 2D plot of the relationship.

Figure 3: Relationship between the first natural frequency, the height h and the complexity of the truss c . The green star indicates the point of maximum first natural frequency, located at $(c = 1, h = 6.6)$ with value $f_1 = 55.39 \text{ Hz}$.

The most stringent design constraint arises from the bridge's natural frequency and drives the final bridge design selection. As shown in Figure 3, increasing the height boosts the first natural frequency until it peaks at $h = 6.6 \text{ m}$, beyond which it declines. Increasing truss complexity past $c = 1$ also decreases the first natural frequency. The optimal configuration, which maximizes the first natural frequency, occurs at $(c = 1, h = 6.6)$ with value $f_1 = 55.39$

Hz. The final bridge layout, including labeled nodes and bar elements, is presented in Figure 4 and detailed in Table 2. The deformation under a 50,000 N load is shown in Figure 7, and the first three mode shapes are illustrated in Figure 6. Key performance metrics and nodal displacements are summarized in Tables 4 and 3, respectively.

Figure 7 shows that the main diagonal members are loaded in compression, while the vertical members are in tension, as is typical with a Howe truss. In this configuration, bars 10 and 12 have symmetric stresses, which are the highest of any member in the truss at 9.46 MPa. However, this remains less than 5% of the yield strength of 304 SS. The maximum node displacement occurs at Node 5 and is only 3.8% of the 40 mm displacement limit. The linear relationship between the applied load and resulting displacement means that the load would need to be scaled by approximately a factor of 35 ($F \approx 1740$ kN) in order for the maximum displacement limit to be reached.

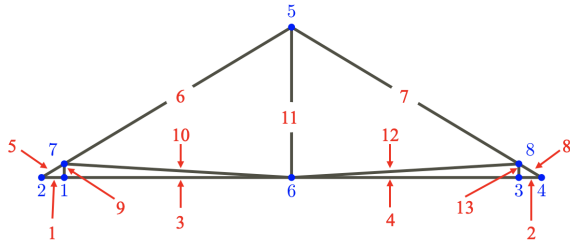


Figure 4: Node and bar numbering for selected design.

Node	x [m]	y [m]
1	0	0
2	-1	0
3	20	0
4	21	0
5	10	6.6
6	10	0
7	0	0.6
8	20	0.6

Table 2: Coordinates of nodes of selected bridge design.

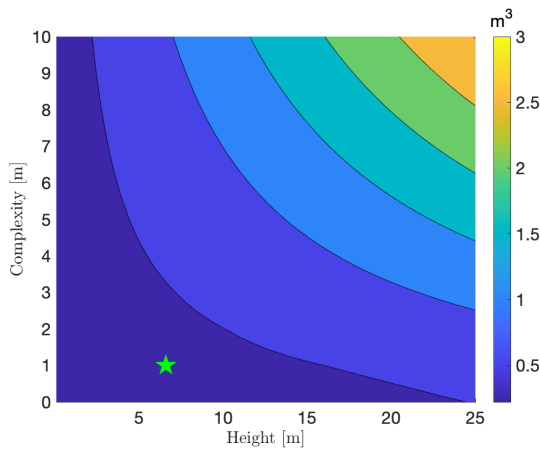


Figure 5: Relationship between structure volume, the height h and the complexity of the truss c . The green star indicates the maximum first natural frequency.

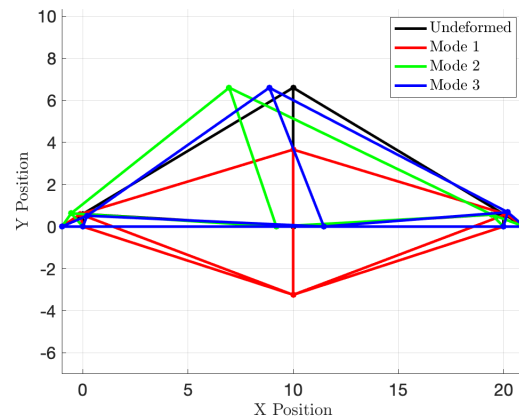


Figure 6: The first three mode shapes for the selected Howe truss.

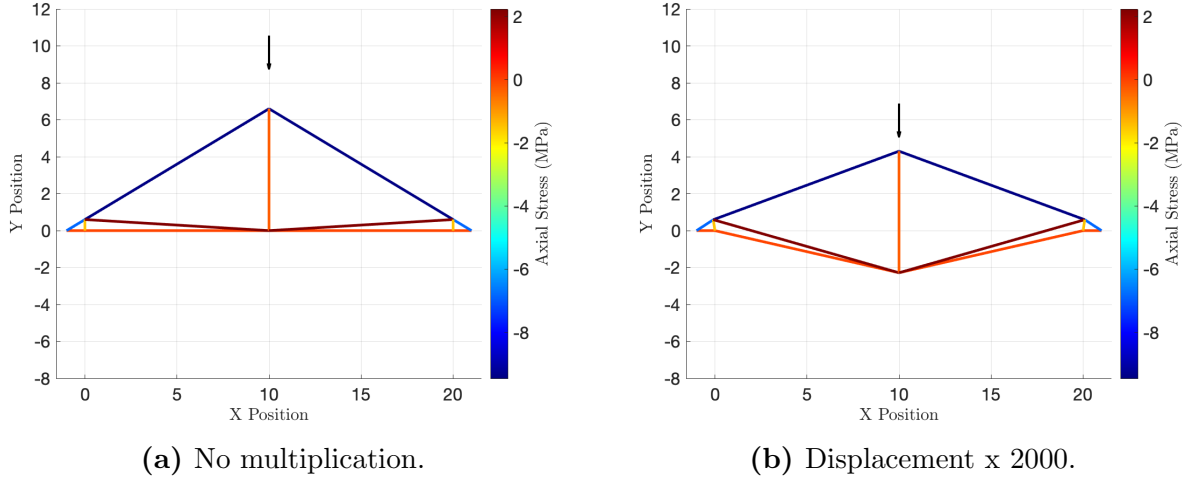


Figure 7: Visualization of the axial stresses across the members of the optimal bridge ($h = 6.6$, $c = 1$) when the 5th node is loaded vertically with 50'000 N. The displacements were multiplied by a factor of 2000 for better visualization (right).

Node	5	6	7	8
displacement [mm]	1.15	1.14	0.04	0.04

Table 3: Norm of the displacements of each node after loading (optimal bridge). The first four nodes are fixed.

Parameter	Value	Target
Volume [m ³]	0.38	< 1.0
Max displacement [mm]	1.15	< 40
Max tensile stress [MPa]	2.24	< 215
Max compressive stress [MPa]	9.46	< 215
1st natural frequency [Hz]	55.39	> 50
2nd natural frequency [Hz]	103.88	~
3rd natural frequency [Hz]	131.67	~

Table 4: Key parameters of selected design.

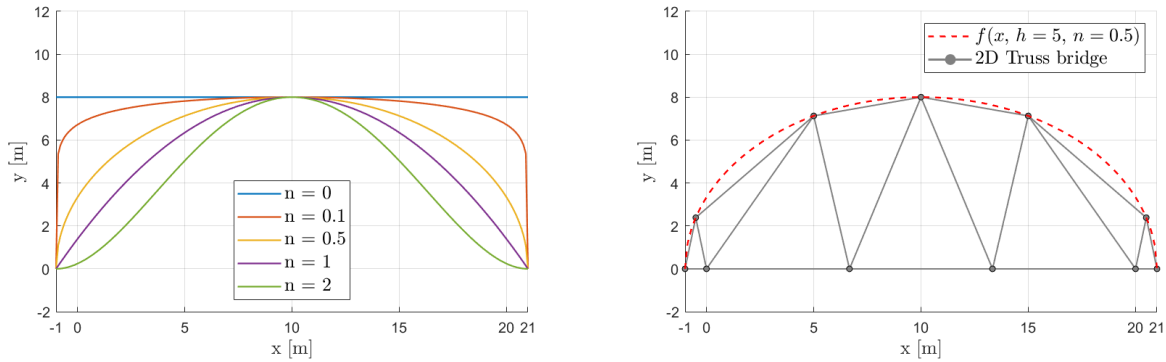
3.2 Case 2: Aluminum alloy

In this case study, the general design philosophy was to combine the Warren and Bowstring truss designs where the curvature of the bridge is controlled by a continuous geometrical variable. This allows more fine tuning to obtain better results regarding the first natural frequency or the volume of material required, for example. In the case of a single loaded node situated at the center of the span of the bridge, a parabola-shaped truss structure is desired to efficiently distribute the load across all members. To describe the outer shape of the bridge and, by extension, the location of the top nodes, a spatial function is needed.

This hypothetical shape function must have two zeros at $(-1, 0)$ and $(21, 0)$, and have a global maximum at the value $x = 10$ (center of the span). To allow more freedom in the definition of the bridge, the height of the fifth node, h , and the form factor, n , are introduced in the shape function. The following function satisfies all the previous requirements.

$$f(x, h, n) = h \left(1 - \left(\frac{x - 10}{11} \right)^2 \right)^n \quad \forall x \in [-1, 21], \forall h \in]0, 10], \forall n \in [0, 2] \quad (13)$$

where the parameters h and n are assigned arbitrary upper bounds, determined based on the observed decline in overall bridge performance beyond those values. The height h cannot be directly equal to 0 because it produces a singular stiffness matrix that leads to an ill-posed system and diverging results.



(a) Shape functions with various n and $h = 8$. (b) 2D truss bridge with $h = 8$ and $n = 0.5$.

Figure 8: Illustration of the shape function f (left) and an example of a bridge with specific h and n parameters (right).

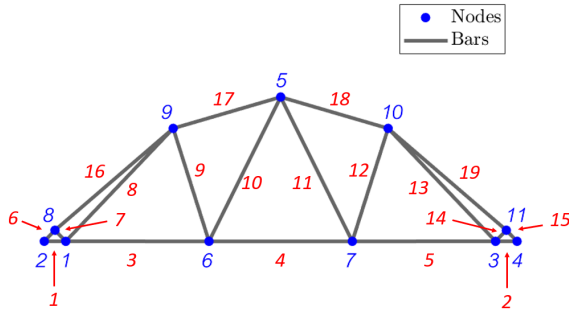
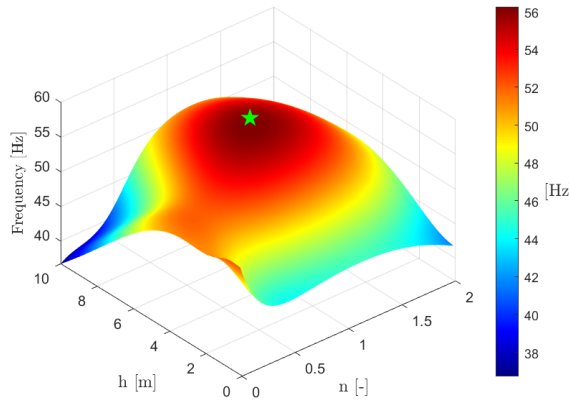


Figure 9: Bridge schematic with numbering.

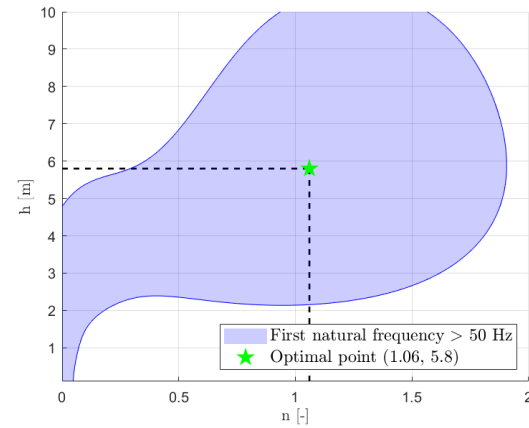
Node	x [m]	y [m]
1	0	0
2	-1	0
3	20	0
4	21	0
5	10	5.8
6	6.67	0
7	13.34	0
8	-0.5	0.45
9	5	4.54
10	15	4.54
11	20.5	0.45

Table 5: Coordinates of nodes.

The coordinates of the nodes excluding the first four ones were chosen specifically to form triangle substructures across the bridge, as they have performed better under stress than rectangular structures, for instance. The shape of the bridge in Figure 9 and coordinates in Table 5 are derived from the optimal value of shape function f in order to obtain the largest first natural frequency. The values of the parameters are $h = 5.8$ m and $n = 1.06$. The x-component of each node are arbitrarily fixed and only the y-component vary according to the shape function (node 6 and 7 are always at the same location between node 1 and 3).



(a) 3D plot of the relationship.



(b) Region where $f_1 > 50$ Hz.

Figure 10: Relationship between the first natural frequency, the height h and the form factor n . The green star indicates the local maximum located at $(n = 1.06, h = 5.8)$ with value $f_1 = 56.25$ Hz.

As observed in Figure 10, the first natural frequency is surprisingly roughly convex with respect to h and n which allows us to find a local maximum. The optimal bridge can then be defined with a unique pair of h and n . This behaviour can be interpreted as the bridge becoming less stiff as h and n deviate too much from the optimal values, thus decreasing f_1 .

In order to satisfy the bonus requirement of having f_1 above 50 Hz, the pair (h, n) needs to lay inside the blue region in Figure 10b.

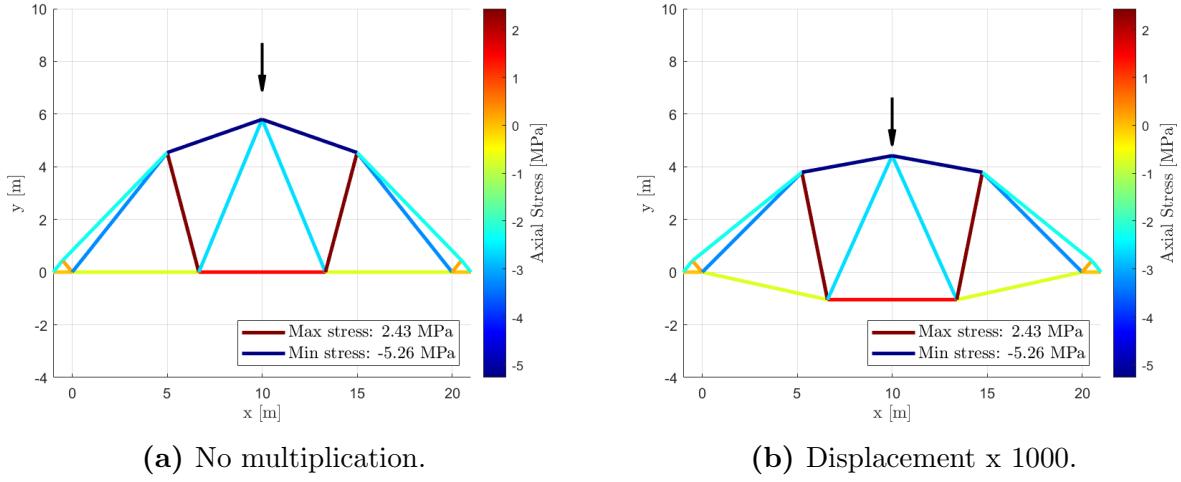


Figure 11: Visualization of the axial stresses across the members of the optimal bridge ($h = 5.8$, $n = 1.06$) when the 5th node is loaded vertically with 50'000 N. The displacements were multiplied by a factor of 1000 for better visualization (right).

When loading the topmost node with a vertical downward force, the bridge experiences generally more compression than traction (see Figure 11). In the optimal bridge's case, the maximum stress does not exceed 1 MPa which is 2 orders of magnitude smaller than the yield strength of 6061 aluminum. Note that the cross-sectional area was taken to be the largest with $A = 70 \text{ cm}^2$. The impact of the cross-sectional area on the total volume of the bridge is shown in Figure 16. As the total volume barely exceeded 1 m^3 in the most extreme case ($h = 10 \text{ m}$ and $n = 0$), A was fixed at 70 cm^2 for the entire case of this study.

Regarding the displacements of each node, all values remain well below the maximum allowable limit of 40 mm (see Table 6). Since the system is analyzed within the linear elastic regime, where the displacement varies linearly with the applied load, it is possible to easily determine the maximum value of the applied force for which the maximum displacement would reach exactly 40 mm. Indeed, an increase of 10% of the applied force results in a 10% increase of the total displacement. This value is $F_{\min} \approx 1450 \text{ kN}$ (about 29 times larger). If F remains constant, we can also find the minimum A which satisfies the maximum displacement requirement. In this case $A_{\min} \approx 2.41 \text{ cm}^2$.

Node	5	6	7	8	9	10	11
displacement [mm]	1.38	1.05	1.05	0.02	0.79	0.79	0.02

Table 6: Norm of the displacements of each node after loading (optimal bridge). The first four nodes are fixed.

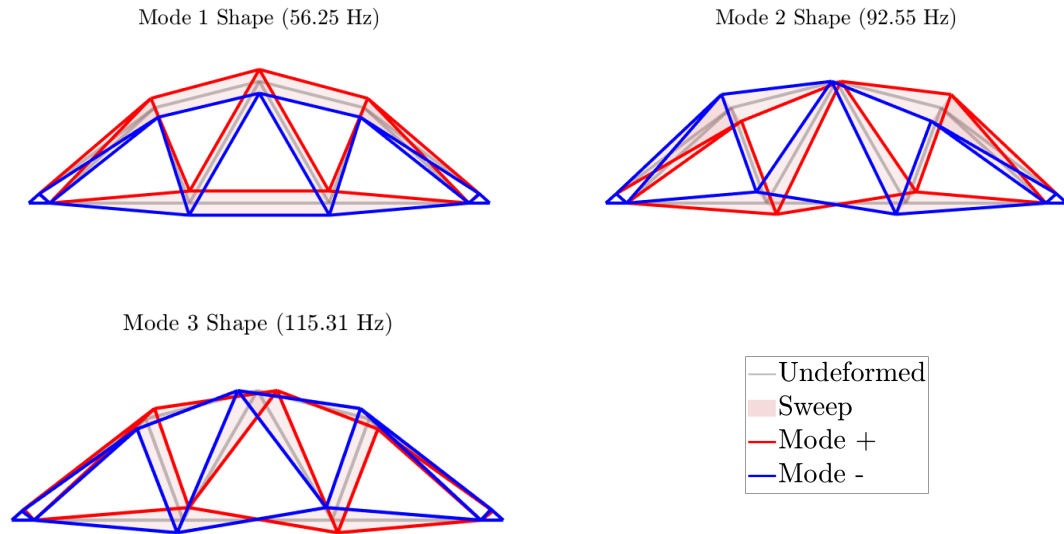


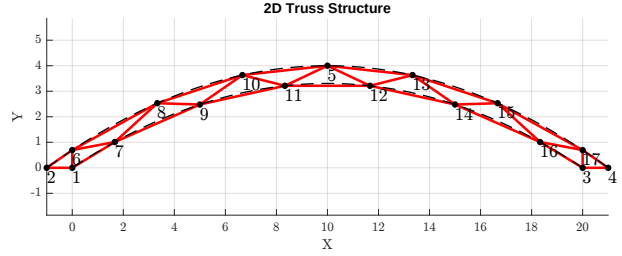
Figure 12: First three mode shapes of the bridge model and their corresponding natural frequencies. Legend explanation: undeformed bridge (grey), sweeping of the bridge during resonance (light red region) between the two extremities of the amplitude $\pm$ (red and blue).

Parameter	Value	Target
Volume [m ³]	0.597	< 1.7
Max displacement [mm]	1.38	< 40
Max tensile stress [MPa]	2.432	< 110
Max compressive stress [MPa]	5.265	< 110
1st natural frequency [Hz]	56.25	> 50
2nd natural frequency [Hz]	92.55	~
3rd natural frequency [Hz]	115.31	~

Table 7: Key parameters of selected design.

3.3 Case 3: Engineered wood

Using the given fixed nodes, a shape of bridge that is fitting them well is an arch bridge. Additionally, this shape brings an alternative shape compared to the first two sections.



(a) As an example, the Sydney harbour bridge. (b) Simplified version used ($a = 4$ m and 6 additional nodes on each sides).

For simplification, the bridge's outer shape consists of 2 identical parabolas that are vertically shifted to fit all the fixed nodes. Then 2 parameters are defined: the height (a) of the node 5 and the number of additional nodes on each side. The minimum of additional nodes on each side was chosen to be 4 due to code limitation, and only even numbers work in this situation. The nodes are equally spaced horizontally and alternate between the 2 shape functions for their vertical position. The maximum cross-sectional area of 100 cm^2 was chosen since it has no impact on the natural frequency and reduces the stress inside the truss while keeping the volume of material below the maximum defined at 2 m^3 . The shape functions are the following:

$$\text{Shape}f_{top}(x) = -\frac{a}{121} \cdot (x+1) \cdot (x-21) \quad \text{Shape}f_{bottom}(x) = \text{Shape}f_{top} - \text{Shape}f_{top}(0) \quad (14)$$

A sweep through the 2 parameters is done. The following results are obtained.

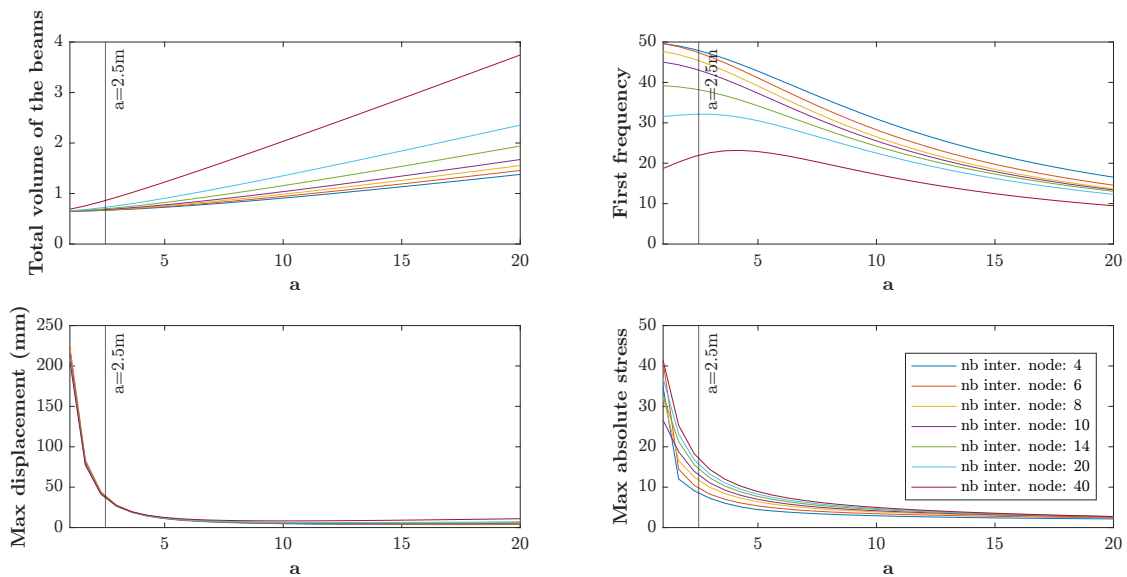


Figure 14: Results of the sweep on the different constraints.

From the various constraints, it appears that a minimum working height of 2.5 m is required to keep node displacements below 40 mm and to avoid excessive stress in the beams. Increasing the height further reduces both displacement and stress, but also decreases the first natural frequency. Interestingly, increasing the number of additional intermediate nodes does not seem to improve any metric. One possible explanation is the truss pattern: in the photo, the inner trusses alternate between vertical and diagonal members, whereas the simulation only used diagonal trusses. Another explanation for this is the length of the elements. Since there is only a point load in the middle of the bridge, making the element smaller means that the elements in the middle of the bridge will have an increased compressive and tension load.

The following final bridge was chosen, with a height of 2.5 m and 4 intermediate nodes on each side:

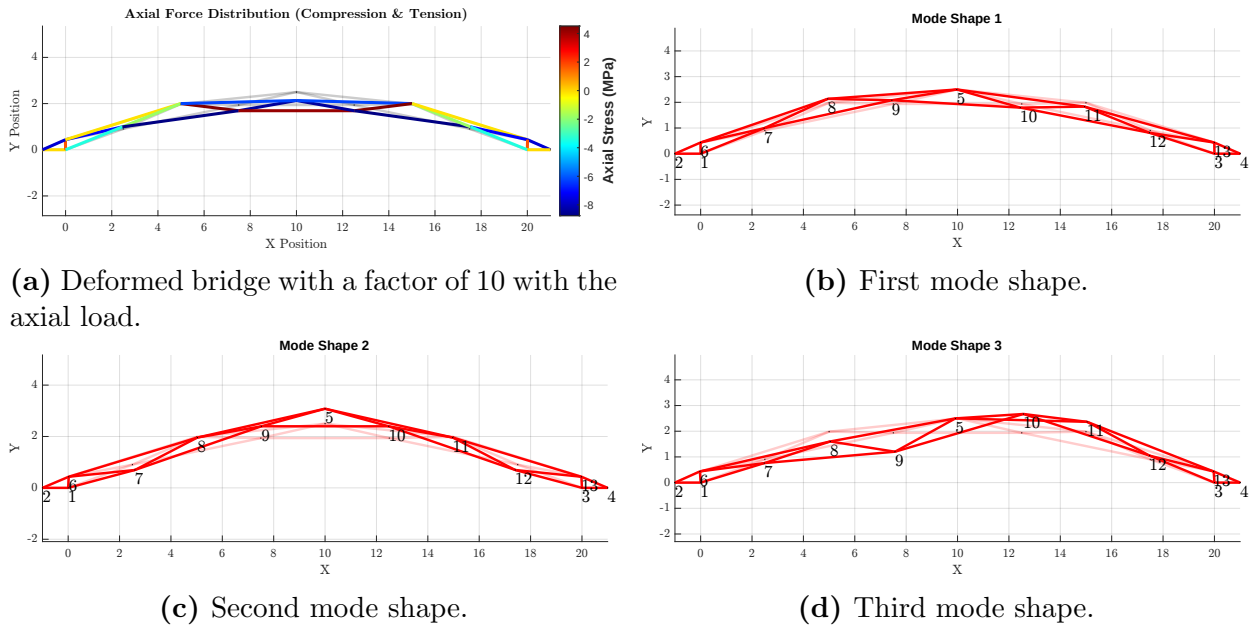


Figure 15: The deformation and mode shapes of the optimal bridge.

Node	5	6	7	8	9	10	11	12	13
displacement [mm]	36.72	0.8	9.61	2.1	24.65	24.65	2.1	9.61	0.8

Table 8: Norm of the displacements of each node after loading (optimal bridge). The first four nodes are fixed.

The bridge trusses are mainly compressed, which is expected with this shape and is advantageous. The total volume is: 0.66758 m^2 which is below the maximum of 2 m^2 given for this material, and the displacement of the nodes is below the 40 mm threshold. However, the fifth approaches the displacement threshold; this may be due to the lower Young's modulus of wood compared to the other materials. Below in Table 9 are the different parameters and their values. The 1st natural frequency fell short of being above 50 Hz. This could

be explained by a larger number of nodes compared to the other sections or by the smaller Young's modulus of engineered wood.

Parameter	Value	Target
Volume [m ³]	0.668	< 2
Max displacement [mm]	36.72	< 40
Max tension stress [MPa]	5.1804	< 30
Max compressive stress [MPa]	8.5039	< 30
1st natural frequency [Hz]	48.18	> 10
2nd natural frequency [Hz]	110.54	~
3rd natural frequency [Hz]	151.11	~

Table 9: Key parameters of selected design.

3.4 Discussion

Across the three bridge designs, several trends can be observed. Firstly, while decreasing the cross-sectional area increases the stress and displacement, it has no impact on the natural frequencies of the structure. This is to be expected, as the mass and stiffness matrices both include the factor A (cross-sectional area), which drops out when solving the eigenvalue problem. Thus, the area was set to be the maximum permitted for each bridge in order to reduce displacements and stress.

Secondly, the first natural frequency of the truss structure has a large dependency on the height of the bridge. All three bridge designs showed that increasing the height up to a certain limit served to increase the natural frequency. This is a result of the higher bending stiffness of the bridge, due to the larger distance between the upper and lower chords. However, past this optimal height, the natural frequency begins to drop. This is a result of the increased length of the bars required to construct the bridge; higher length lowers the natural frequency, and the improvements in bending stiffness can no longer overcome this change in mass. Another way to explain this, is to notice that a $1/\ell^2$ term remains in the term $\omega = \sqrt{\mathbf{K}/\mathbf{M}}$ when computing the natural frequencies. As h increases, the overall ℓ length of the bars increase as well and thus $\omega \propto 1/\ell^2 \rightarrow 0$.

Thirdly, while stress is dependent on the bridge shape (including number of nodes, overall form, and connectivity), the bridge height has a much higher impact on maximum stress. For example, for a given bridge height of 2.5 m, the steel bridge maximum stress varies only from 67 MPa to 75 MPa for complexities from 1 to 10 (a 9% change). However, changing the bridge height from 2.5 m to 0.25 m increases the maximum stress from 75 MPa to 367 MPa for the same shape (a 389% increase). In this case, the height much more quickly increases the general bending stress of the bridge than adding nodes or interior trusses, meaning the aspect ratio is the most important factor. Overall, increasing the bridge height

reduces the maximum stress in the truss. As the height increases, the bending stiffness also increases (as discussed previously), resulting in lower stress. In a similar way, as bridge height approaches zero, the maximum stress in the truss diverges (approaches infinity). This is due to the fundamental assumption that the members of the truss can only take axial loads. When the bridge height is zero, all members are horizontal and are perpendicular to the applied load. Thus, they cannot efficiently support the vertical load, requiring an infinitely high internal stress.

Overall, despite differences in material and form, all three bridges achieved a maximum natural frequency of approximately 50 Hz, with maximum stresses less than 10 MPa. Looking more closely at the designs, the parabolic form chosen for the aluminum bridge is optimized to reduce stress and the first natural frequency, while the structural steel Howe design has the lowest displacement. The wooden truss design has significantly higher displacements, which may be attributed to its lower young's modulus and potentially also to its lack of more vertically oriented members. Both studies that investigated the effects of added nodes/internal members found optimal results for lower complexities. While adding more members and nodes can help reduce stress, it reduces the first natural frequency since it increases the degrees of freedom of the bridge. For example, at the optimal height of 2.5 m for the wood arch bridge, when using 4 additional nodes on each side or 40 additional nodes, the first natural frequency is divided by 2. For the steel Howe bridge, also at optimal height of 6.6 m, the first natural frequency is divided by 2 when increasing from 8 nodes to 20.

Using only a truss approach (considering only axial forces) instead of a frame approach (considering axial forces and bending moments at the nodes) makes calculations much easier; however, it deviates from reality, especially if displacement becomes large. Since no moments at the node are considered, nodes can freely rotate. If displacements become large, this assumption will not hold anymore as the moments at the nodes will be significant. Therefore, the truss approach may not be well suited to the wooden bridge design in this case, as it has quite large displacements.

4 Conclusion

This project investigated the design and optimization of three 2D truss bridges using finite element methods, revealing that bridge height has the greatest influence on natural frequency and stress, while shape and internal node arrangement refine performance within limits. Although increasing internal nodes adds degrees of freedom, it does not always improve metrics and can lower natural frequencies if poorly distributed. The study confirms that an optimal balance of simplicity, height, and shape is more effective than purely increasing complexity, and highlights how pure truss models, while simplifying calculations, introduce assumptions like neglecting moments and the rigid connections on the pins, explaining possible deviations from real-world behavior. As noted in the introduction, the use of frictionless pinned joints and ideal truss elements neglects bending moments and connection

effects, greatly simplifying the analysis but causing slight deviations from real-world bridge behavior. Additionally, while important metrics like member buckling could have been explored, they were omitted in this study as stresses remained well within yield limits. Future improvements in the design could include reducing the cross-sectional areas of the beams at locations where stress is low to optimize the overall volume.

5 Appendix

5.1 Case 2: Aluminum alloy

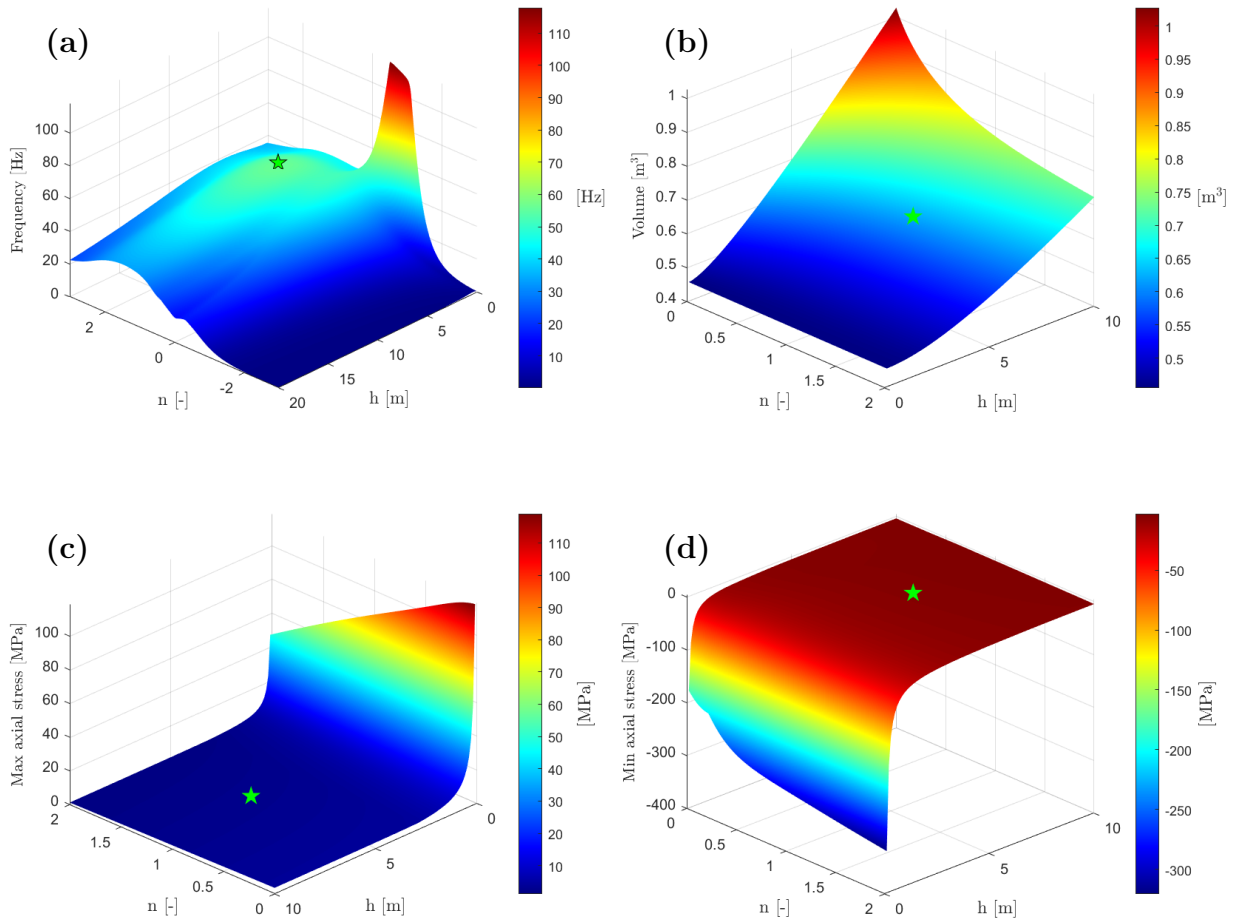


Figure 16: **(a)** First natural frequency with an extended (h, n) domain ($h \in]0, 20]$ and $n \in [-3, 3]$). We can observe the singularity around the point $(h = 0, n = -0.5)$. **(b)** Total volume of aluminum alloy used in the bridge with respect to h and n . The shape factor n affects more the volume as h increases. The volume of the optimal bridge is $V = 0.597 \text{ m}^3$. **(c)** Maximum axial stress across all members. The stress tends to ∞ as $h \rightarrow 0$. This is possibly due to the truss system becoming collinear, leading to a loss of vertical load-carrying capacity through axial forces alone. Consequently, maintaining equilibrium necessitates unbounded internal axial stresses, resulting in their divergence. **(d)** Minimum axial stress across all members. The stress diverges to $-\infty$ as $h \rightarrow 0$.